Chapter 8 Notes

Section 1

Notes

Here is why, for any class of ordinals B, the smallest ordinal $\alpha \in B$ is an initial element of B If $\alpha = \emptyset$, then $\alpha \cap B = \emptyset$. If $\alpha \neq \emptyset$, then all members of α are ordinals less than α. But, by assumption, there are no ordinals in B less than α, meaning $\alpha \cap B = \emptyset$.

Exercise 1.1

(1) We first show that R_Ω is well-founded. Consider any non-empty subclass A of R_Ω. Then let α be the lowest rank that any of the elements in A have. Now consider any element $x \in A$ which has rank α. Suppose that there was some element y of A in x. Then y would have rank $\geq \alpha$. But this is not possible, as, by P_6 of Chapter 7 (and the fact that x is a set), y would have a rank $< \alpha$, a contradiction. Thus all elements of x are not in y, meaning $x \cap A = \emptyset$. Thus every element $x \in A$ with rank α is an initial element of A.

(2) The fact that every R_α (where $\alpha > 0$) is well-founded follows because every R_α is a non-empty subclass of R_Ω. Obviously R_0 is also well-founded (because $\emptyset$ is well-founded).

If now it was the case that R_Ω was in fact equal to V, this exercise would prove that V was well-founded.

Theorem 2.3

Here is a more formal proof of Theorem 2.3 (a more formal form of the second paragraph of the proof).

Since B is non-empty, it contains an element e. We now form the denumerable sequence $b_1, b_2, ..., b_n, ...$ as follows: $b_1 = e$; if b_n is an initial element of B, then $b_{n+1} = b_n$; if b_n is not an initial element of B, then we take b_{n+1} to be the least element (under the well-ordering of A) of b_n that is in B.

As remarked before the proof of Theorem 2.3, A satisfying the descending $\in$-chain condition means that A contains no $\in$-cycles. If all elements of the above denumerable sequence are distinct, then clearly that sequence is an infinite descending $\in$-chain, contradicting the assumption that A satisfies the descending $\in$-chain condition.

Suppose, on the other hand, that two elements of the sequence were the same (say the elements b_m and b_n, where $m < n$), and that b_m was not an initial element of B. Then, for all i where $m \leq i < n$, it must be that $b_i \in b_{i+1}$. But that means we would have an $\in$-cycle $b_m, b_{m+1}, ..., b_n$, which cannot happen.

Thus it cannot be that all elements of the sequence are distinct, or that two elements of the sequence were the same where one of them was not an initial element of B. Hence it must be that a pair of elements of the sequence are the same, where one of the elements of the pair is an initial element.

Section 3

Notes

It is remarked that an infinite descending chain of ordinals cannot exist. This is because an infinite descending chain of ordinals would not have a least element, contrary to the fact that every class of ordinals has a least element.

Theorem 3.6

The proof of this theorem does not seem correct to me. It says: "Either V_0 contains all ordinals of V or it does not. Suppose it does. Then $R_\Omega \subseteq V_0$." This is not correct, as it was not assumed that V contains all ordinals. Thus V_0 containing all ordinals of V does not necessarily imply that $R_\Omega \subseteq V_0$ (because R_Ω contains all ordinals). This shouldn't be an issue, as the case where On is not a subclass of V can then be proven in the way starting with "If, on the other hand, V_0 does not contain all ordinals of V..." in the proof.

However, the part of the proof starting with that sentence seems to be more complicated than the proof lets on. It claims that the rank of every element of V_0 must be in V_0, which does require proof.

Proof. By the induction method of Theorem 3.2. Suppose the hypothesis holds for the ranks of all elements of some element x in V_0. Then the rank α of x is the least ordinal that is greater than all the ranks of the elements of x. Either α is a successor ordinal or α is a limit ordinal. Let O be the set of all ranks of elements of x. Note that $O \in V_0$ by the axiom of substitution (applied in V_0). Then $\cup O$ is an ordinal and is in V_0 (because V_0 is a Zermelo-Fraenkel universe). If α is a successor ordinal, then $\alpha = \cup O \cup \{\cup O\}$, which is also in V_0. If α is a limit ordinal, then $\cup O = \alpha$. In any case, $\alpha \in V_0$, therefore completing the proof. □

Thus the rank of every element of V_0 is less than α (the α defined in the proof). It is then concluded that this means that $V_0 \subseteq R_\alpha$. This is because all elements of V_0 have rank less than α, meaning they are all members of R_α.

Exercise 3.1

We prove that, for any Zermelo-Fraenkel universe V, the intersection of all Zermelo sub-universes of V, each containing all the ordinals of V, is well founded.

Proof. If V contains all ordinals, then the above intersection is R_Ω (by Exercise 4.1 of Chapter 7), which is well founded (by Exercise 1.1). On the other hand, suppose that V does not contain all ordinals. Then it can easily be seen that the least ordinal that V does not contain must be some limit ordinal λ. We aim to show that $R_\lambda \subseteq V$. First, we show that for every $\alpha < \lambda$, $R_\alpha \in V$. Obviously $R_0 \in V$. Suppose some $R_\alpha \in V$. Then $\mathcal{P}(R_\alpha) \in V$ (because V is a Zermelo-Fraenkel universe), that is, $R_{\alpha+1} \in V$. On the other hand, suppose that for all ordinals α less than some limit ordinal λ_1 such that $\lambda_1 < \lambda$, $R_\alpha \in V$. Then $R_{\lambda_1} = \cup_{\alpha<\lambda_1} R_\alpha$. Because $\lambda_1 < \lambda$, and V contains all ordinals less than λ, $\lambda_1 \in V$. Thus the set of all R_α for all $\alpha < \lambda_1$ is in V (by the axiom of substitution), meaning so is $\cup_{\alpha<\lambda} R_\alpha$, that is, R_{λ_1}. Hence, for all $\alpha < \lambda$, $R_\alpha \in V$. By the transitivity of V, it then obviously follows that $R_\lambda \subseteq V$.

Now, R_λ is a Zermelo universe (as shown in Chapter 7), and contains all ordinals of V. Also R_λ is well founded. Thus the intersection of all Zermelo sub-universes of V, each containing all the ordinals of V, is a subclass of R_λ, meaning it is well founded (because all subclasses of a well founded class are well founded). □

Exercise 3.2

This exercise does not seem correct to me. For example, consider the set $R_3 = \{0, \{0\}, \{1\}, \{0, 1\}\}$. It is well-founded (by Exercise 1.1) and transitive, so it satisfies the assumptions of this exercise. Then we could let G be the set of all ordered pairs $\langle x, \{0, 1\}\rangle$ where $x \in R_3$. Now, suppose we had a function F satisfying the conditions of the

exercise. Then $F(0) = G(F''(0)) = G(0) = \{0, 1\}$ and thus (because $x \cap R_3 = x$ for any $x \in R_3$, as R_3 is transitive) $F(1) = G(F''(1)) = G(\{\{0, 1\}\})$. But G is not defined for $\{\{0, 1\}\}$, which means this is an apparent contradiction.